

Zhubing Jia, Ph. D.

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SUMMARY

Experimental quantum physicist with 9+ years of hands-on experience across trapped-ion and neutral atom tweezer array platforms. Deep expertise in quantum control, optical system design, and numerical simulation of quantum system operations. Proven track record of high-impact publications in *Nature Physics*, *PRX Quantum* and *PRL*, and a US patent in optomechanical system design.

TECHNICAL SKILLS

Quantum platforms	Neutral atom tweezer arrays (^{171}Yb), trapped-ion systems ($^{171}\text{Yb}^+$, Ca^+/K)
Quantum control	Pulse engineering, robust gate design
Simulation & modeling	Open-system simulation, noise modeling, QEC circuit-level simulation (small codes, custom implementation)
Experimental Programming	UHV systems, cryogenic ion trap systems, atom-photon entanglement Python (NumPy, SciPy, QuTiP, Matplotlib, etc)

RESEARCH EXPERIENCE

University of Illinois Urbana-Champaign 2023 - present

- Developed and benchmarked coherent qubit operations on ^{171}Yb neutral atom tweezer arrays (up to 40 atoms); routinely operated 20-qubit register for quantum simulation and networking experiments
- Achieved 95% fidelity atom-photon entanglement between a neutral atom qubit and telecom-band photon, enabling scalable quantum network node architecture
- Led numerical simulation and modeling efforts for all major experiments using Python/QuTiP; developed open quantum system models incorporating realistic noise and decoherence

Duke University 2018 - 2022

- Contributed to design and characterization of a high-stability cryogenic ion trap system
- Designed and implemented frequency-modulated two-qubit gate pulse sequences for $^{171}\text{Yb}^+$ ion chains, improving angle robustness and raising two-qubit gate fidelity from 97.84% to 98.11% under realistic noise
- Developed multi-mode motional state tomography protocol for trapped-ion chains; demonstrated full quantum state reconstruction of motional modes
- Demonstrated quantum simulation of conical intersections – a classically intractable molecular dynamics problem – using trapped ions

Georgia Institute of Technology 2016 - 2017

- Co-built a hybrid UHV trap co-trapping Ca^+ ions and K atoms; characterized atom-ion interactions via mass spectrometry

SELECTED PUBLICATIONS

- Lintao Li, Xiye Hu, **Zhubing Jia**, William Huie, *et al.* Parallelized telecom quantum networking with a ytterbium-171 atom array, *Nature Physics*, 21, pp. 1826–1833 (2025).

- **Zhubing Jia**, William Huie, Lintao Li, *et al.* An Architecture for Two-qubit Encoding in Neutral Ytterbium-171 Atoms, *npj Quantum Information* 10, 106 (2024).
- William Huie, Lintao Li, Neville Chen, Xiye Hu, **Zhubing Jia**, *et al.* Repetitive Readout and Real-time Control of Nuclear Spin Qubits in ^{171}Yb Atoms, *PRX Quantum* 4, 030337 (2023).
- Jacob Whitlow, **Zhubing Jia**, Ye Wang, *et al.* Quantum Simulation of Conical Intersections using Trapped Ions, *Nature Chemistry* 15, pp. 1509–1514 (2023).
- **Zhubing Jia**, Shilin Huang, Mingyu Kang, *et al.* Angle-robust Two-Qubit Gates in a Linear Ion Crystal, *Phys. Rev. A* 107, 032617 (2023).
- **Zhubing Jia**, Ye Wang, Bichen Zhang, *et al.* Determination of Multi-mode Motional Quantum States in a Trapped Ion System, *Phys. Rev. Lett.* 129, 103602 (2022).
- Robert Fulton Spivey, Ismail Volkan Inlek, **Zhubing Jia**, *et al.* High-Stability Cryogenic System for Quantum Computing with Compact Packaged Ion Traps, *IEEE Transactions on Quantum Engineering*, vol. 3, pp. 1-11 (2022).
- S. Jyothi, Kisra N. Egodapitiya, Brad Bondurant, **Zhubing Jia**, *et al.* A hybrid ion-atom trap with integrated high resolution mass spectrometer, *Review of Scientific Instruments* 90, 103201 (2019).

NEWS ARTICLES

- **Zhubing Jia**, Quantum logic with bosonic error correction, *Nature Physics*, 21, pp. 1528–1529 (2025).

PATENTS

- Robert Spivey, Kai Hudek, Ismail Inlek, Jungsang Kim, **Zhubing Jia**. High-Stability Optomechanical System and Cryo-package Assembly for Quantum Computing, *US Patent Application 17883062* (2023).

EDUCATION

2018 - 2022	Ph. D. in physics (transferred from Georgia tech), Duke University	Durham, NC
2016 - 2017	Ph. D. in physics (coursework), Georgia Institute of Technology	Atlanta, GA
2015 - 2016	Visiting student, King's College London	London, UK
2012 - 2016	B. Sc in physics, Wuhan University	Wuhan, China

AWARDS AND HONORS

IQUIST Postdoctoral Fellowship

01/2023

University of Illinois Urbana-Champaign

ACADEMIC SERVICES

Journal referee

Nature, Nature Physics, Science Advances, PRX Quantum, Physical Review Letters, etc.